

1. Least Squares: Three Perspectives

Let $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. The parameters $\mathbf{x} \in \mathbb{R}^n$ produce predictions $A\mathbf{x}$. Ordinary least squares minimizes the sum of squared residuals:

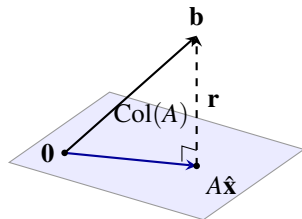
$$\hat{\mathbf{x}} \in \operatorname{argmin}_{\mathbf{x}} \|\mathbf{b} - A\mathbf{x}\|^2, \quad \hat{\mathbf{b}} = A\hat{\mathbf{x}}, \quad \mathbf{r} = \mathbf{b} - \hat{\mathbf{b}}.$$

Fitting observations. For a line $b_i \approx sa_i + c$, each row describes one observation. Matrix-vector multiplication collects the predictions:

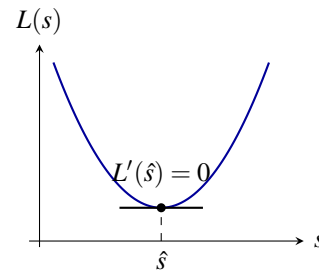
$$\underbrace{\begin{bmatrix} a_1 & 1 \\ a_2 & 1 \\ \vdots & \vdots \\ a_m & 1 \end{bmatrix}}_{\text{model matrix } A} \underbrace{\begin{bmatrix} s \\ c \end{bmatrix}}_{\text{parameters } \mathbf{x}} = \underbrace{\begin{bmatrix} sa_1 + c \\ sa_2 + c \\ \vdots \\ sa_m + c \end{bmatrix}}_{\text{predictions } A\mathbf{x}} \approx \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}}_{\text{observations } \mathbf{b}}.$$

The objective adds the squares of the vertical residuals $b_i - (sa_i + c)$ in the data plot.

Projection. The best prediction is the point in $\operatorname{Col}(A)$ closest to $\mathbf{b}$. The residual is perpendicular to this space. Here is a plane in $\mathbb{R}^3$:



Calculus. With one slope parameter, $L(s) = \|\mathbf{b} - s\mathbf{a}\|^2$ is a quadratic. For $\mathbf{a} \neq \mathbf{0}$, its minimum occurs where $L'(\hat{s}) = 0$:



In several parameters, setting each partial derivative to zero gives the same residual orthogonality condition as projection: the **normal equations**

$$A^\top (\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0} \iff A^\top A\hat{\mathbf{x}} = A^\top \mathbf{b}.$$

If the columns of A are linearly independent, the unique minimizing parameters are $\hat{\mathbf{x}} = (A^\top A)^{-1} A^\top \mathbf{b}$.

2. Building a Classifier

We would like to develop a classifier to classify points based on their distance from the origin. You are presented with the following data. Each data point $\mathbf{d}_i^T = [x_i \ y_i]^T$ has the corresponding label $l_i \in \{-1, 1\}$. For each model below, we will consider the **sign** of the output to determine if the predicted label is -1 or 1 .

x_i	y_i	l_i
-2	1	-1
-1	1	1
1	1	1
2	1	-1

Labels for data you are classifying

- (a) You want to build a model to understand the data. You first consider a linear model, i.e. you want to find $\alpha, \beta, \gamma \in \mathbb{R}$ such that $l_i \approx \alpha x_i + \beta y_i + \gamma$. Set up a least squares problem to solve for α, β , and γ . If this problem is solvable, solve it, i.e. find the best values for α, β, γ . If it is not solvable, justify why.

- (b) You now consider a model with a quadratic term: $l_i \approx \alpha x_i + \beta x_i^2$ with $\alpha, \beta \in \mathbb{R}$. Set up a least squares problem to fit the model to the data. If this problem is solvable, solve it, i.e, find the best values for α, β . If it is not solvable, justify why.

x_i	y_i	l_i
-2	1	-1
-1	1	1
1	1	1
2	1	-1

Labels for data you are classifying

- (c) Finally, you consider the model: $l_i \approx \alpha x_i + \beta x_i^2 + \gamma$, where $\alpha, \beta, \gamma \in \mathbb{R}$. Independent of the work you have done so far, **would you expect this model or the model in part (b) (i.e. $l_i \approx \alpha x_i + \beta x_i^2$) to have a smaller error in fitting the data? Explain why.**

3. Least Squares with Orthogonal Columns

Suppose we would like to solve the least squares problem for $\mathbf{A} \in \mathbb{R}^{3 \times 2}$ and $\mathbf{b} \in \mathbb{R}^3$; that is, find an optimal vector $\mathbf{x} \in \mathbb{R}^2$ which gets $\mathbf{Ax}$ closest to $\mathbf{b}$ such that the distance $\|\mathbf{e}\| = \|\mathbf{b} - \mathbf{Ax}\|$ is minimized. Call this optimal vector $\hat{\mathbf{x}} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix}$. Mathematically, we can express this as:

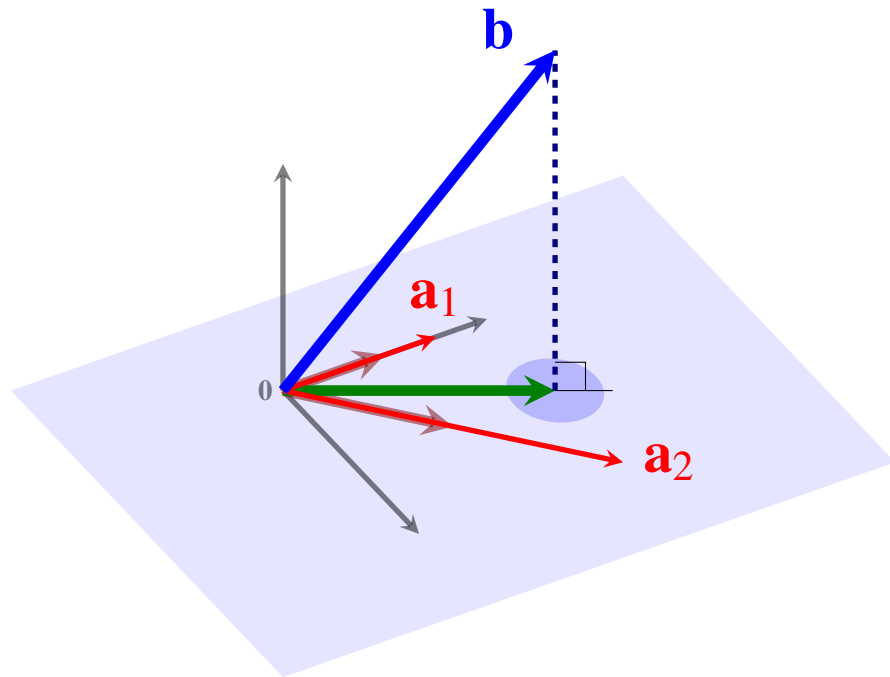
$$\|\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}\|^2 = \min_{\mathbf{x} \in \mathbb{R}^2} \|\mathbf{b} - \mathbf{Ax}\|^2 = \min_{\mathbf{x} \in \mathbb{R}^2} \left\| \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} - \begin{bmatrix} | & | \\ \mathbf{a}_1 & \mathbf{a}_2 \\ | & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|^2.$$

To identify the solution $\hat{\mathbf{x}}$, we may recall the least squares formula: $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$, which is applicable when $\mathbf{A}$ has linearly independent columns. We would now like to walk through the intuition behind this formula for the case when $\mathbf{A}$ has orthogonal columns: $\langle \mathbf{a}_1, \mathbf{a}_2 \rangle = 0$.

(a) On the diagram below, please label the following elements:

*NOTE: For this sub-part only, the matrix $\mathbf{A}$ does **not** have orthogonal columns.*

$\text{span}\{\mathbf{a}_1, \mathbf{a}_2\}$ $\mathbf{A}\hat{\mathbf{x}}$ $\hat{x}_1 \mathbf{a}_1$ $\hat{x}_2 \mathbf{a}_2$ $C(\mathbf{A})$ $\mathbf{e} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$ $\text{proj}_{C(\mathbf{A})}(\mathbf{b})$.

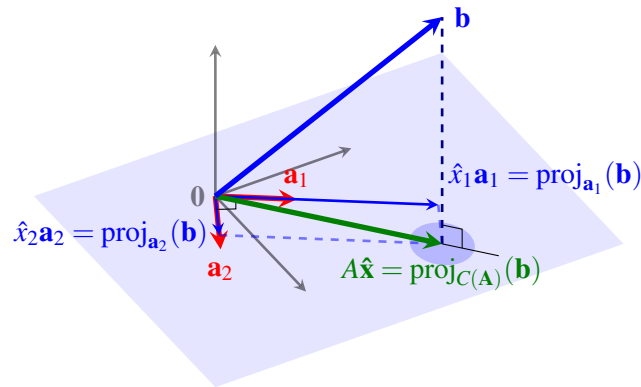


- (b) Now suppose we assume a special case of the least squares problem where the columns of $\mathbf{A}$ are orthogonal (illustrated in the figure below). Given that $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$, and $\text{proj}_{C(\mathbf{A})}(\mathbf{b}) = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b} = \mathbf{A}\hat{\mathbf{x}}$, show the following statement holds:

$$\langle \mathbf{a}_1, \mathbf{a}_2 \rangle = 0 \implies \hat{\mathbf{x}} = \begin{bmatrix} \frac{\langle \mathbf{a}_1, \mathbf{b} \rangle}{\|\mathbf{a}_1\|^2} \\ \frac{\langle \mathbf{a}_2, \mathbf{b} \rangle}{\|\mathbf{a}_2\|^2} \end{bmatrix} \quad \text{and} \quad \text{proj}_{C(\mathbf{A})}(\mathbf{b}) = \text{proj}_{\mathbf{a}_1}(\mathbf{b}) + \text{proj}_{\mathbf{a}_2}(\mathbf{b})$$

In words, the statement says that when the columns of $\mathbf{A}$ are orthogonal, the entries of the least squares solution vector $\hat{\mathbf{x}}$ can be computed by using $\mathbf{b}$ and only the single other vector $\mathbf{a}_i$, and that the projection of $\mathbf{b}$ onto $C(\mathbf{A})$ can be computed by summing the projections of $\mathbf{b}$ onto the $\mathbf{a}_i$.

RECALL... $\text{proj}_{\mathbf{a}_1}(\mathbf{b}) = \frac{\langle \mathbf{a}_1, \mathbf{b} \rangle}{\|\mathbf{a}_1\|^2} \mathbf{a}_1$, $\|\mathbf{a}\|^2 = \langle \mathbf{a}, \mathbf{a} \rangle$ $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ $\mathbf{A} = \begin{bmatrix} | & | \\ \mathbf{a}_1 & \mathbf{a}_2 \\ | & | \end{bmatrix}$



- (c) Compute the least squares solution $\hat{\mathbf{x}} \in \mathbb{R}^2$ to the following system:

$$\min_{\mathbf{x} \in \mathbb{R}^2} \left\| \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|^2.$$

HINT: Notice that the columns of $\mathbf{A}$ are orthogonal!